

# Cyclone Case Study

The reconstruction resolves a cyclone's cold wake - once you ask the question the right way.

| PORT | PAGE FILE                         | STORM                          | RESULT                           |
|------|-----------------------------------|--------------------------------|----------------------------------|
| 8517 | app/phase2/cyclone_volume_page.py | SHAKHTI, Arabian Sea, Oct 2025 | cooling at 11 of 12 track points |

Every other feature in this pack shows the model matching something it was trained toward. This one shows it reproducing a physical process that appears nowhere in its training objective - and that is a stronger statement than any error figure.

## 1. What this feature is, in plain words

When a cyclone crosses the ocean it mixes and upwells cold water from below, leaving a **cold wake** along its track that stays visible for a week or more. Our model is driven only by surface satellite fields and was never taught that cyclones cool the ocean beneath them. If the reconstruction shows that wake anyway, it is reproducing real physics rather than reciting a fit.

It does. Along cyclone SHAKHTI's Arabian Sea track in October 2025, the cyclone heat potential fell at **11 of 12 track points - 92%**, with a mean drop of 4.26 kJ/cm<sup>2</sup>.

**But the first way we measured it said there was no wake at all, and that half is just as interesting.** Comparing one fixed 'before' date against one fixed 'after' date gave a mean change of -0.18 kJ/cm<sup>2</sup> and cooling at only 7 of 12 points. Same storm, same reconstruction, same code. Only the question changed.

The reason is physical. A storm takes days to cross a basin, so one pair of dates asks the wrong question at every point but one: water at 68 east was hit on day one and had already recovered by the 'after' date, while water at 60 east was hit on day six and its wake was three days old. Comparing each point against **its own passage time** is the right question. **The page shows both**, because showing only the good number would be asking to be trusted.

### THE EVERYDAY PICTURE

Photograph a crowd before and after a parade passes and you will conclude nothing happened - the front of the route has already emptied while the back has not filled yet. Photograph each spot as the parade reaches it, and the pattern is unmistakable.

## 2. What you actually see on the screen

One storm, two measurements, and a three-dimensional object that is honestly labelled:

- The storm track with the change in heat potential marked at every track point.
- **Both** measurements side by side - the naive fixed-date pair, and the passage-relative one - so the contrast is the point rather than a footnote.
- A verdict in words, not just a number, so the page cannot render a null result under the word 'wake'.
- An isosurface of the **warm layer itself** - the body of water above 26 degrees, whose thickness is D26 - thinning along the track. That is the fuel being spent.
- Both agencies' peak wind estimates, with their disagreement stated rather than resolved in favour of the larger one.

## 3. How to run it, step by step

#### RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/cyclone_volume_page.py --server.port 8517
```

- 1 Run the command and open **http://localhost:8517**.
- 2 Select cyclone SHAKHTI, Arabian Sea, 1-7 October 2025.
- 3 **Read the cost warning before you start the run.** A week-long track needs 14 whole-basin reconstructions - about seven minutes on CPU, two on GPU. The page states this before it spends it, because a page that goes quiet for seven minutes looks broken.
- 4 Show the naive fixed-date result first. It reports essentially nothing.
- 5 Then show the passage-relative result: cooling at 11 of 12 points, mean -4.26 kJ/cm<sup>2</sup>. Let the contrast land.
- 6 Finish on the warm-layer isosurface thinning along the track - this is the picture people remember.

#### A DIMENSIONAL POINT THE PAGE INSISTS ON

'A 3-D heat potential volume' is dimensionally confused: heat potential is a depth *integral*, measured in kJ per square centimetre, and a 2-D field has no volume. So this page does not draw one. What is three-dimensional, and is exactly what a cyclone eats, is the warm layer itself - and the caption says which is which.

## 4. Why the problem statement needs this

The problem statement is a disaster-management problem statement. Demonstrating that the reconstruction captures a real cyclone-ocean interaction is the most direct evidence that it is fit for that purpose - far more so than an aggregate error figure.

It is also the strongest scientific validation in the pack. Matching a training target shows a model has fitted. Reproducing an untaught physical process, from surface data alone, shows it has learned something about how the ocean works.

| PS requirement                               | How this feature answers it                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Proof of concept over the Arabian Sea</b> | A real named storm, a real track, a real measured ocean response.                                             |
| <b>Daily temporal resolution</b>             | Passage-relative measurement is only possible with daily fields - a monthly average erases the entire signal. |
| <b>Disaster-management relevance</b>         | The cold wake governs whether a following storm finds fuel, and is directly operational.                      |

## 5. The measured numbers behind it

The measurement, and the measurement that would have been wrong [VERIFIED]:

| How the question was asked                     | Mean change                                   | Points that cooled    |
|------------------------------------------------|-----------------------------------------------|-----------------------|
| <b>Each point against its own passage time</b> | <b>-4.26 kJ/cm<sup>2</sup></b> , largest -7.4 | <b>11 of 12 (92%)</b> |
| <b>One fixed before/after date pair</b>        | -0.18 kJ/cm <sup>2</sup>                      | 7 of 12 (58%)         |

- Same storm, same reconstruction, same code. **Only the question changed.** The function returns a verdict *string* as well as a number, so a page structurally cannot render -0.18 under the word 'wake', and a test asserts that a flat field yields '*no clear cold wake*'.

- **A correction we made about our own reporting.** An earlier internal note quoted the storm's peak wind as 74 knots. The archive carries two agency estimates that disagree: one gives a peak of 60 knots over 41 rated points, the other 74 knots over 29 - **14 knots apart, spanning a storm-category boundary**. The loader now carries both, prefers the internationally-endorsed figure, and exposes the disagreement rather than quietly picking the larger number.
- **An error of our own, caught by rendering the page.** One panel reported that a date was 'outside the bundle'. It was not - the date simply had not been reconstructed yet. Reporting one absence as a different absence, on a page whose whole subject is measuring absences correctly. The two cases are now distinguished.

## 6. The impact - and who it is for

| Who                        | What changes for them                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Cyclone forecasters</b> | Evidence that the reconstruction tracks the ocean's actual response to a storm, which is what intensity forecasting depends on. |
| <b>Disaster management</b> | A basis for assessing whether a second storm following a first will find depleted or replenished fuel.                          |
| <b>Ocean scientists</b>    | Daily basin-wide cold-wake analysis from satellites, for storms where no float happened to be in the way.                       |
| <b>The jury</b>            | The clearest evidence in the pack that the model learned physics rather than memorising a target.                               |

### WHY SOCIETY GAINS FROM THIS

Cyclones in the North Indian Ocean kill and displace people on a scale few other natural hazards match in this region. A system that demonstrably captures the ocean's response to a storm - daily, everywhere, from satellites - contributes directly to the forecasting chain that decides when a coastal district is told to move.

## 7. Say this to the jury

*This is the result I would most like you to remember. A cyclone drags cold water up behind it, leaving a cold wake. Our model is driven only by surface satellite fields and was never taught that cyclones cool the ocean. Along SHAKHTI's track, heat potential fell at 11 of 12 points, mean 4.26 kJ per square centimetre. But look at the first measurement: comparing one fixed before-and-after pair, we found essentially nothing - seven of twelve, mean 0.18. Same storm, same code. Only the question changed, because a storm takes days to cross a basin and one date pair asks the wrong question everywhere but one point. We show you both, because showing only the good number would be asking to be trusted.*

Close the demo with this. It is the strongest scientific claim in the project.

## 8. Questions the jury will ask

### Q. Could the cold wake simply be in the satellite input already?

**A.** The surface cooling is, and that is honest - the model reads sea-surface temperature. What the model is not told is how that surface signal propagates into a subsurface heat content change along a moving track. Reproducing the depth-integrated response at 11 of 12 points, from surface fields alone, is the result.

### Q. Is one storm enough to prove anything?

**A.** No, and we do not claim it proves a general capability. It is a case study on one named storm, and we present it as one. What it does establish is that the mechanism is present in the reconstruction and that we know how to measure it correctly - including which way of measuring it would have produced a false negative.

**Q. Why does your naive measurement disagree with your headline one?**

**A.** Because a storm takes days to cross the basin, so a single before-and-after pair asks the wrong question at every point but one. That is a physical fact about moving storms, not a choice of favourable numbers - which is exactly why both results are on the page and the function returns a verdict in words as well as a number.

## 9. The honest limit - say it before they find it

### ONE STORM, ONE BASIN, ONE WEEK

This is a case study, not a validated capability. No cold-wake climatology was computed, no second storm was analysed, and the Bay of Bengal counterpart is specified but not run. The measurement also inherits every limit of the underlying heat-content product, including the model's warm bias - and the cost is real: 14 whole-basin reconstructions, about seven minutes on CPU, which the page states before it spends it.

#### IN ONE LINE

*The reconstruction reproduces a cyclone's cold wake it was never taught - and the measurement that would have found nothing is shown beside it.*

**11 of 12**

TRACK POINTS COOLED, PASSAGE-RELATIVE

**-4.26 kJ/cm2**

MEAN DROP ALONG THE TRACK

**7 of 12**

WHAT THE NAIVE QUESTION FOUND INSTEAD

Feature 17 of 17. Next: the closing note - conclusion, limitations and viva questions.